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PROGRAMMING WITH PYTHON

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Computer Applications

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Python Basics and Standard Library

Functions

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- A function is a block of organized, reusable code that is used to perform a single, related action.
- It is a named section of a program that performs a specific task.
- Functions provide better modularity for the application and a high degree of code reusing.

- Function blocks begin with the keyword **def** followed by the function name and parentheses **( )**.
  - The first statement of a function can be an optional statement - the documentation string of the function or docstring.
  - The code block within every function starts with a colon (:) and is indented.

- Any input parameters or arguments should be placed within these parentheses.
- The statement return [expression] exits a function, optionally passing back an expression to the caller.
  - A return statement with no arguments is the same as return None.

- By default, parameters have a positional behavior and have to be informed in the same order that they were defined.
  - def functionname(parameters):
  - "function_docstring"
  - function_suite
  - return [expression]

Calling Functions

- Defining a function only gives it a name, specifies the parameters that are to be included in the function and structures the blocks of code.
- Once the basic structure of a function is finalized, execute it by calling it from another function or directly from the Python prompt.

Pass by reference vs. Pass by value

- Pass by reference
  - It is used in some programming languages, where address of the variable is passed and then the operation is done on the value stored at these addresses.

- Pass by value
  - It means that the value is directly passed as the value to the argument of the function.
  - Here, the operation is done on the value and then the value is stored at the address.
  - Pass by value is used for a copy of the variable.

Pass by reference vs. Pass by value

- All parameters (arguments) in the Python language are passed by reference.
- If what a parameter refers to within a function is changed, the change also reflects back in the calling function.
- The parameter has no specified type.
- The type is decided when the argument is copied on call to the function at runtime – type is determined dynamically.

Arguments

- Required arguments
- Keyword arguments
- Default arguments
- Variable-length arguments

Required Arguments

- Required arguments are the arguments passed to a function in correct positional order.

- The number of arguments in the function call should match exactly with the function definition

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Required Arguments

```
def printme( name, age ):
    "This prints a passed string into this function"
    print (name , age)
```

```
printme("Ajay",30)
```

```
Ajay 30
```

```
printme("Ajay")
```

```
---------------------------------------------------------------------------
TypeError                 Traceback (most recent call last)
Cell In[3], line 1
----> 1 printme("Ajay")

TypeError: printme() missing 1 required positional argument: 'age'
```

Keyword Arguments

- Keyword arguments are related to the function calls.
- When keyword arguments are used in a function call, the caller identifies the arguments by the parameter name.

Keyword Arguments

```
def printme( name, age ):
    "This prints a passed string into this function"
    print (name , age)
```

```
printme(age=30, name="Ajay")
```

```
Ajay 30
```

```
printme(name="Ajay")
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[9], line 1
----> 1 printme(name="Ajay")

TypeError: printme() missing 1 required positional argument: 'age'
```

Default Arguments

- A default argument is an argument that assumes a default value if a value is not provided in the function call for that argument.

```
def printme( name, age=40 ):
    "This prints a passed string into this function"
    print (name , age)
```

```
printme("Ajay")
```

```
Ajay 40
```

```
printme("Ajay", 30)
```

```
Ajay 30
```

Variable-Length Arguments

- A function may process more arguments than specified number of arguments.
- These arguments are called variable-length arguments and are not named in the function definition,
- An asterisk (*) is placed before the variable name that will hold the values of all non-keyword variable arguments.

- This tuple remains empty if no additional arguments are specified during the function call.
- def functionname ([formal_args,] *var_args_tuple):
- "function_docstring"
- function_suite
- return [expression]

Variable-Length Arguments

```
def multiplier(*num):
    prod = 1
    for i in num:
        prod = prod * i
    print("Product:",prod)
```

```
multiplier(3,5)
```

```
Product: 15
```

```
multiplier(1,2,4)
```

```
Product: 8
```

```
multiplier(2,2,6,7)
```

```
Product: 168
```

Recap

- Functions

- Arguments to a function

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THANK YOU

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